EmbedUR Networks Training

Assessment – 4

Module – 7: Layer 3 – Ports and SNMP Protocol

&

Module – 8: Transport Layer Fundamentals

Name: Kishore S

1. Try Test-Connection and nslookup commands for below websites

www.google.com

www.facebook.com

www.amazon.com

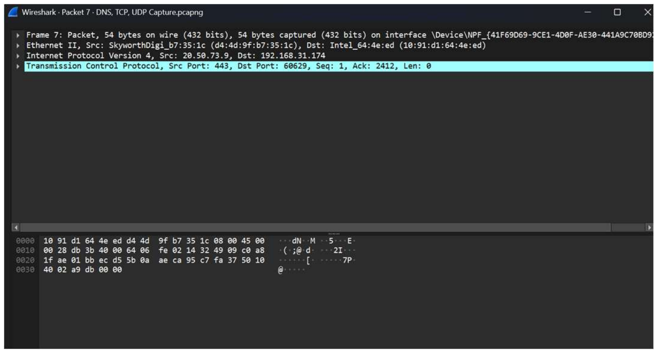
www.github.com

www.cisco.com

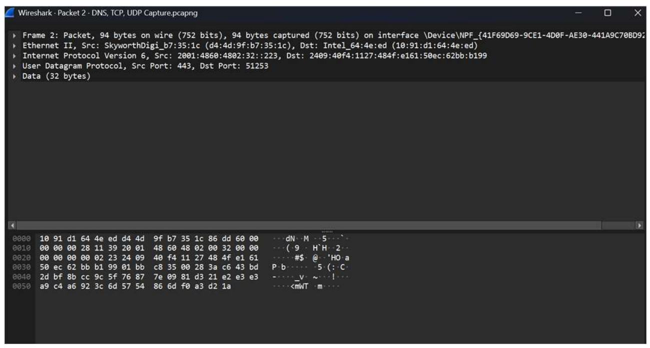
```
kishore@kishore-virtual-machine:~$ nslookup google.com
Server:         127.0.0.53
Address:        127.0.0.53#53

Non-authoritative answer:
Name:   google.com
Address: 142.251.221.110
Name:   google.com
Address: 2404:6800:4007:815::200e

kishore@kishore-virtual-machine:~$ nslookup facebook.com
Server:         127.0.0.53
Address:        127.0.0.53#53

Non-authoritative answer:
Name:   facebook.com
Address: 57.144.210.1
Name:   facebook.com
Address: 2a03:2880:f369:1:face:b00c:0:25de

kishore@kishore-virtual-machine:~$ nslookup github.com
Server:         127.0.0.53
Address:        127.0.0.53#53

Non-authoritative answer:
Name:   github.com
Address: 20.207.73.82

kishore@kishore-virtual-machine:~$ nslookup cisco.com
Server:         127.0.0.53
Address:        127.0.0.53#53

Non-authoritative answer:
Name:   cisco.com
Address: 72.163.4.185
Name:   cisco.com
Address: 2001:420:1101:1::185

kishore@kishore-virtual-machine:~$ nslookup amazon.com
Server:         127.0.0.53
Address:        127.0.0.53#53

Non-authoritative answer:
Name:   amazon.com
Address: 98.82.161.185
Name:   amazon.com
Address: 98.87.170.74
```

2. Use Wireshark to capture and analyze DNS, TCP, UDP traffic and packet header, packet flow, options and flags.

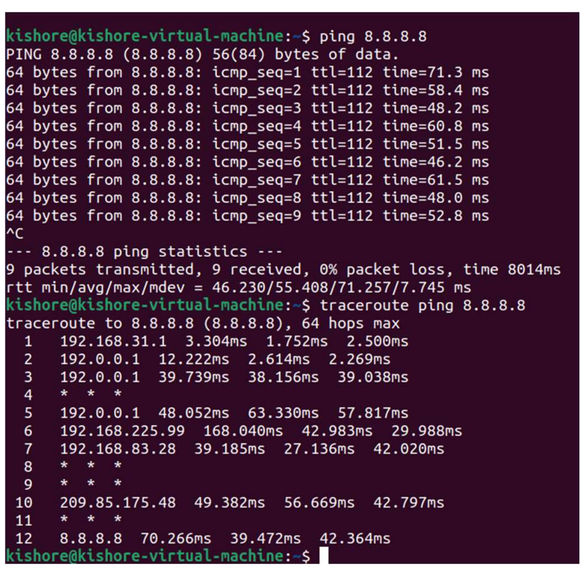
TCP:



UDP:



3. Explore traceroute/tracert for different websites eg:google.com and analyse the parameters in the output and explore different options for traceroute command



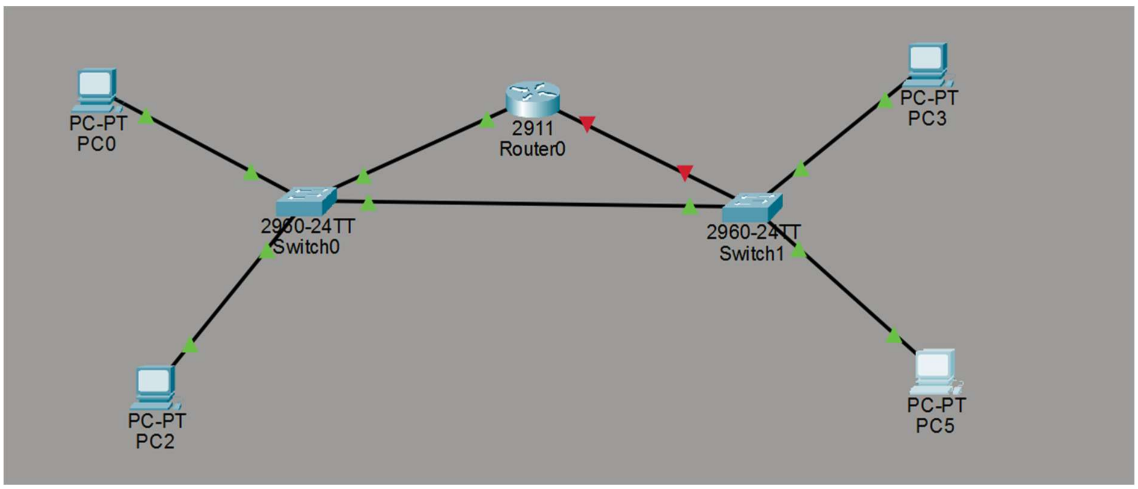
4. Use Cisco packet tracer for the below

Set up trunk ports between switches and try ping between different VLANs.

Change the native VLAN on a trunk port.Test for VLAN mismatches and troubleshoot.

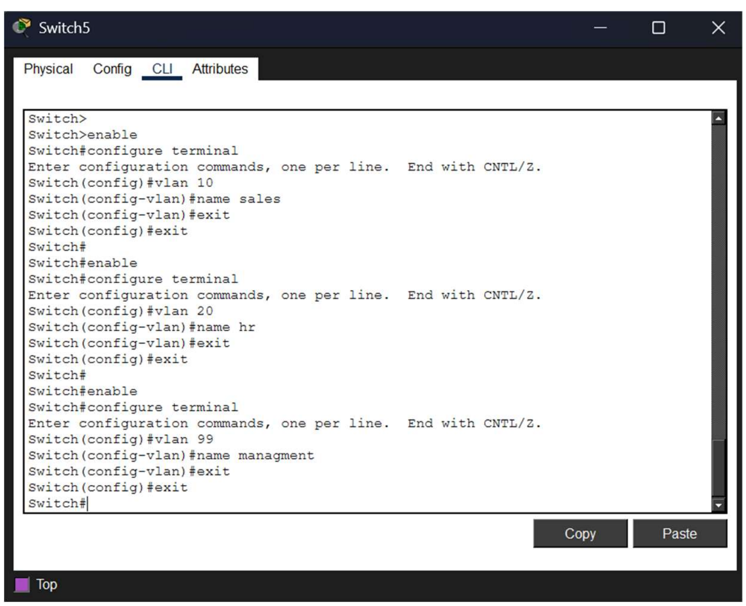
Configure a management VLAN and assign an IP address for remote access. Test SSH or Telnet access to the switch.

Network Topology:



Switch - 1 Config:

Set VLAN Identifier Names:

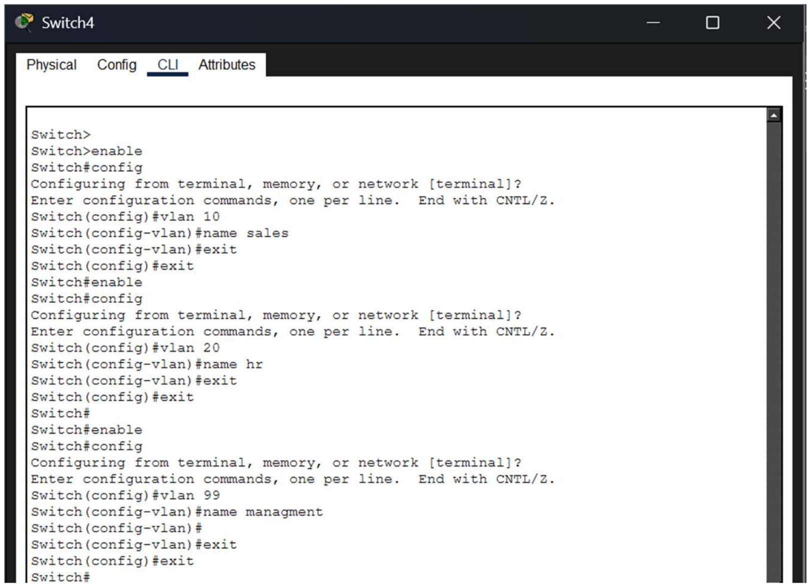


Set Port Types for every VLANs:

```
Switch>enable
Switch#config
Configuring from terminal, memory, or network [terminal]?
Enter configuration commands, one per line.  End with CNTL/Z.
Switch(config)#
Switch(config)#interface fa0/1
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 10
Switch(config-if)#exit
Switch(config)#exit
Switch#
Switch#enable
Switch#config
Configuring from terminal, memory, or network [terminal]?
Enter configuration commands, one per line.  End with CNTL/Z.
Switch(config)#interface fa0/2
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 20
Switch(config-if)#
Switch(config-if)#exit
Switch(config)#exit
Switch#
Switch#enable
Switch#config
Configuring from terminal, memory, or network [terminal]?
Enter configuration commands, one per line.  End with CNTL/Z.
Switch(config)#interface fa0/24
Switch(config-if)#exit
Switch(config)#exit
Switch#
Switch#enable
Switch#config
Configuring from terminal, memory, or network [terminal]?
Enter configuration commands, one per line.  End with CNTL/Z.
Switch(config)#interface fa0/2
Switch(config-if)#exit
Switch(config)#exit
Switch#
Switch#enable
Switch#config
Configuring from terminal, memory, or network [terminal]?
Enter configuration commands, one per line.  End with CNTL/Z.
Switch(config)#interface fa0/3
Switch(config-if)#switchport mode trunk

Switch(config-if)#switchport trunk native vlan 99
Switch(config-if)#exit
Switch(config)#exit
Switch#
```

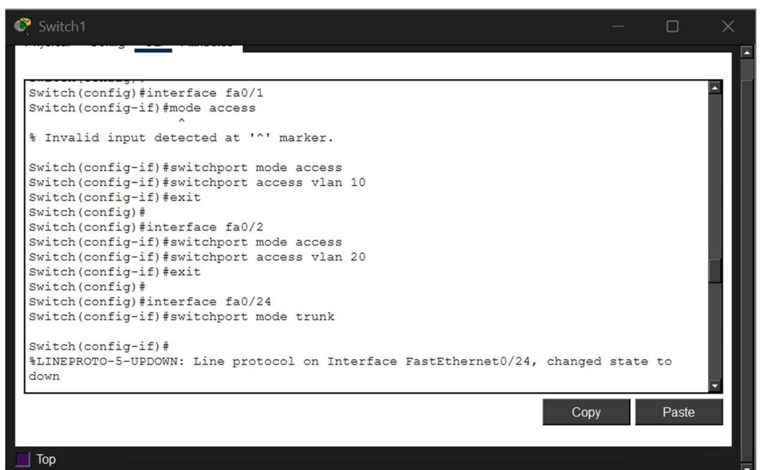
Switch -2 Config:

Create VLAN Identifiers:

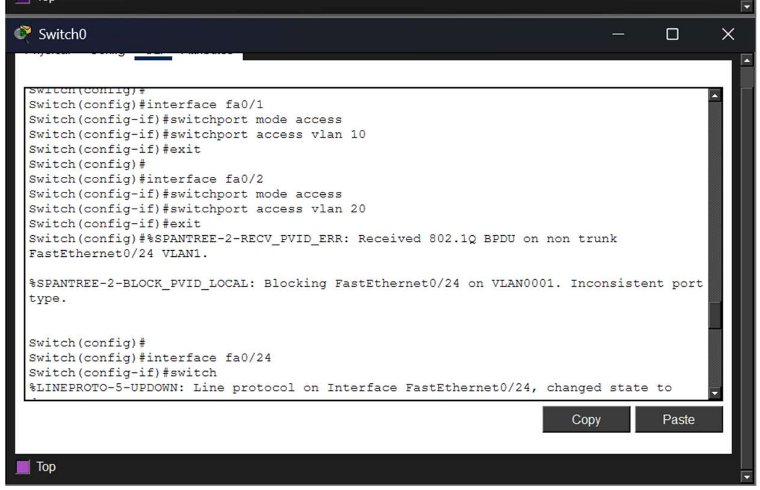



```
Switch>
Switch>enable
Switch#config
Configuring from terminal, memory, or network [terminal]?
Enter configuration commands, one per line.  End with CNTL/Z.
Switch(config)#vlan 10
Switch(config-vlan)#name sales
Switch(config-vlan)#exit
Switch(config)#exit
Switch#enable
Switch#config
Configuring from terminal, memory, or network [terminal]?
Enter configuration commands, one per line.  End with CNTL/Z.
Switch(config)#vlan 20
Switch(config-vlan)#name hr
Switch(config-vlan)#exit
Switch(config)#exit
Switch#
Switch#enable
Switch#config
Configuring from terminal, memory, or network [terminal]?
Enter configuration commands, one per line.  End with CNTL/Z.
Switch(config)#vlan 99
Switch(config-vlan)#name managment
Switch(config-vlan)#
Switch(config-vlan)#exit
Switch(config)#exit
Switch#
```

Assign Access Ports:

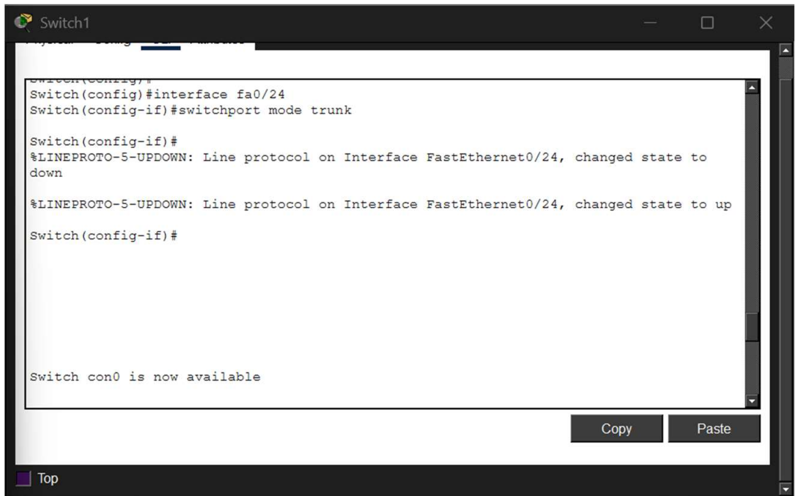


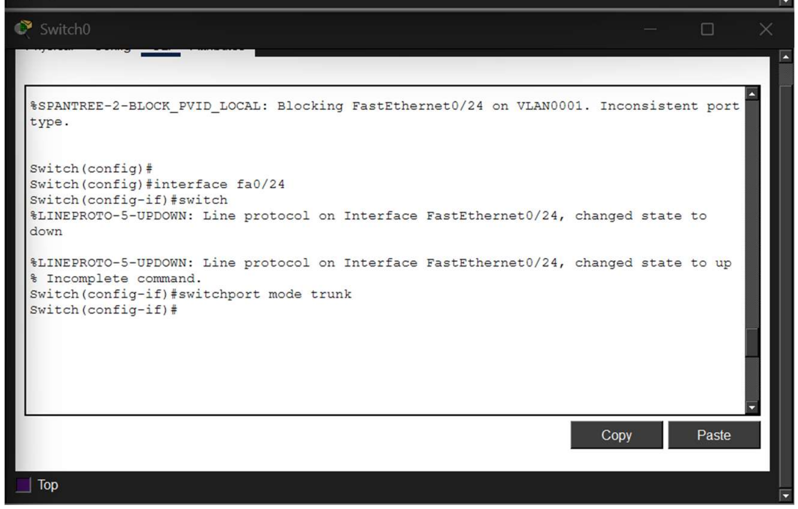

```
Switch(config)#interface fa0/1
Switch(config-if)#mode access
                    ^
% Invalid input detected at '^' marker.

Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 10
Switch(config-if)#exit
Switch(config)#
Switch(config)#interface fa0/2
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 20
Switch(config-if)#exit
Switch(config)#
Switch(config)#interface fa0/24
Switch(config-if)#switchport mode trunk

Switch(config-if)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/24, changed state to
down
```



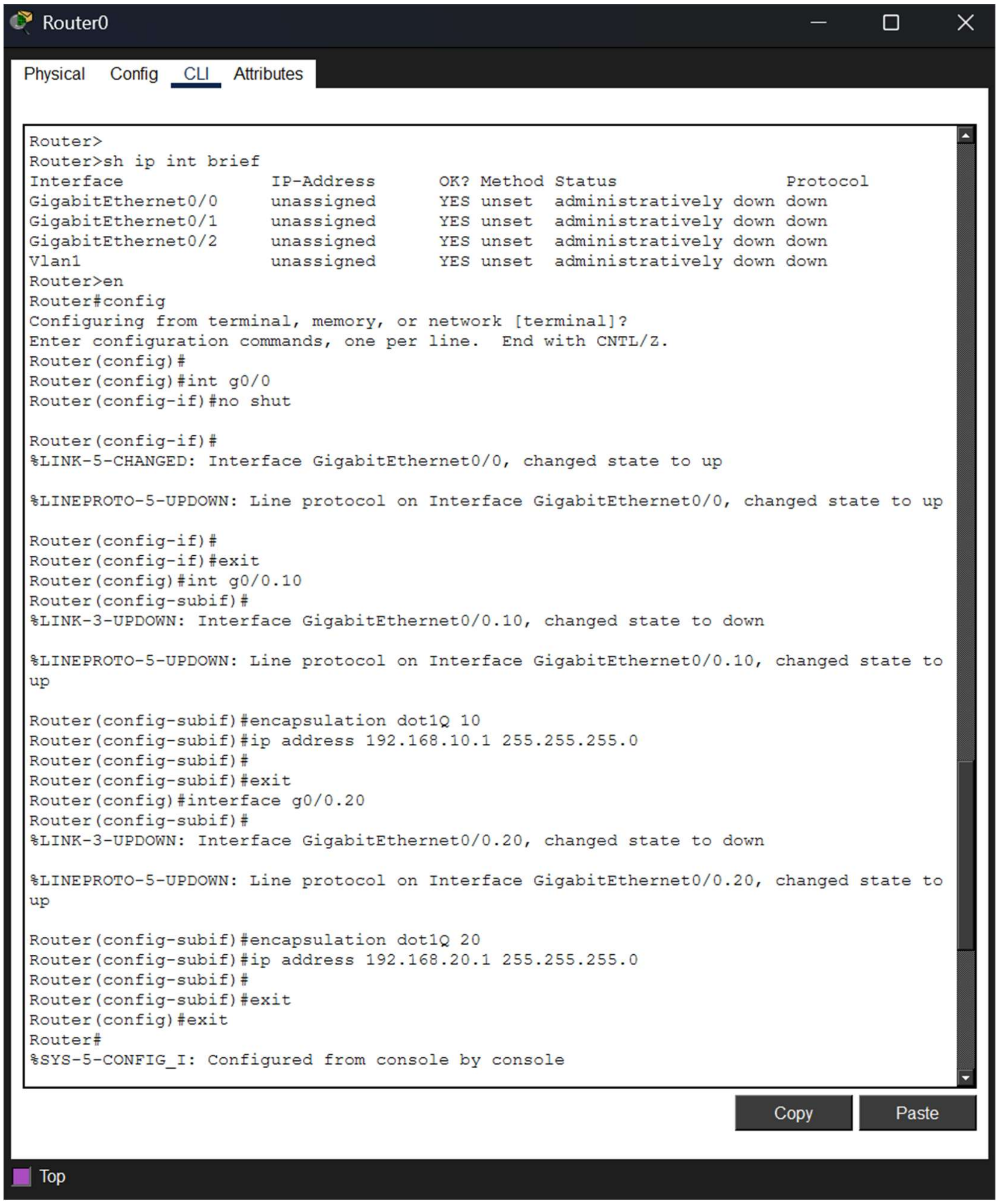

```
Switch(config)#interface fa0/1
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 10
Switch(config-if)#exit
Switch(config)#
Switch(config)#interface fa0/2
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 20
Switch(config-if)#exit
Switch(config)#%SPANTREE-2-RECV_PVID_ERR: Received 802.1Q BPDU on non trunk
FastEthernet0/24 VLAN1.

%SPANTREE-2-BLOCK_PVID_LOCAL: Blocking FastEthernet0/24 on VLAN0001. Inconsistent port
type.


Switch(config)#
Switch(config)#interface fa0/24
Switch(config-if)#switch
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/24, changed state to
```

Assign Trunk Ports:





Router Config:

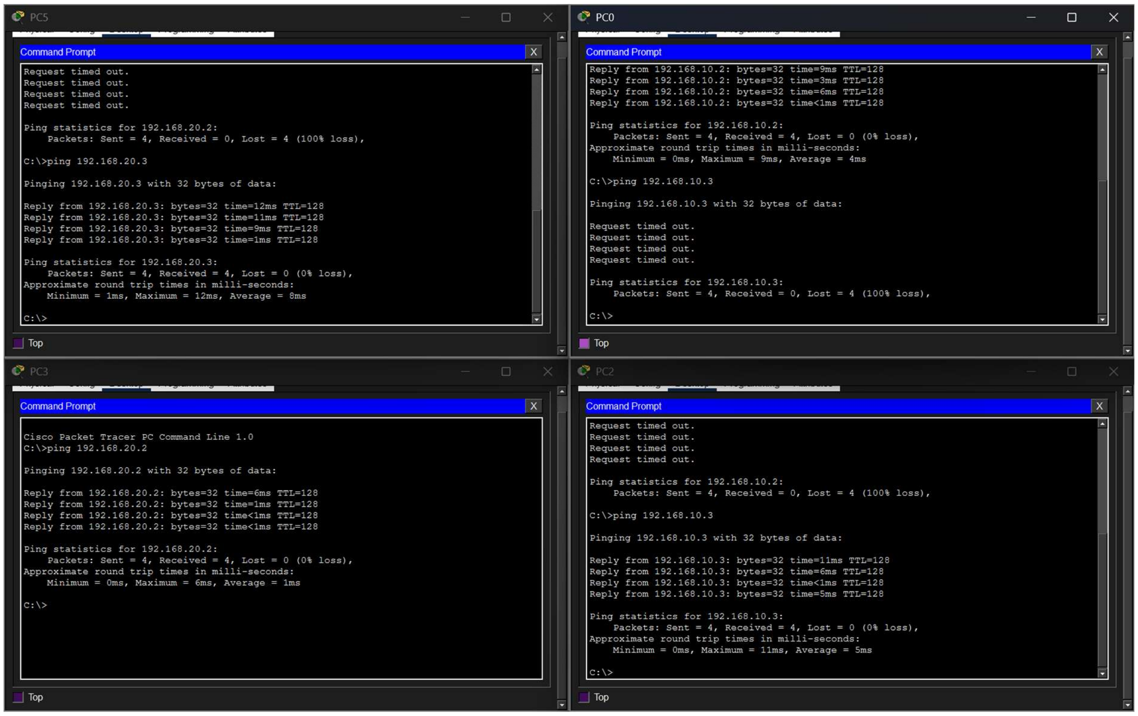



```
Router>
Router>sh ip int brief
Interface              IP-Address      OK? Method Status                Protocol
GigabitEthernet0/0     unassigned      YES unset  administratively down down
GigabitEthernet0/1     unassigned      YES unset  administratively down down
GigabitEthernet0/2     unassigned      YES unset  administratively down down
Vlan1                  unassigned      YES unset  administratively down down
Router>en
Router#config
Configuring from terminal, memory, or network [terminal]?
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#
Router(config)#int g0/0
Router(config-if)#no shut

Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up

Router(config-if)#
Router(config-if)#exit
Router(config)#int g0/0.10
Router(config-subif)#
%LINK-3-UPDOWN: Interface GigabitEthernet0/0.10, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0.10, changed state to
up

Router(config-subif)#encapsulation dot1Q 10
Router(config-subif)#ip address 192.168.10.1 255.255.255.0
Router(config-subif)#
Router(config-subif)#exit
Router(config)#interface g0/0.20
Router(config-subif)#
%LINK-3-UPDOWN: Interface GigabitEthernet0/0.20, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0.20, changed state to
up

Router(config-subif)#encapsulation dot1Q 20
Router(config-subif)#ip address 192.168.20.1 255.255.255.0
Router(config-subif)#
Router(config-subif)#exit
Router(config)#exit
Router#
%SYS-5-CONFIG_I: Configured from console by console
```

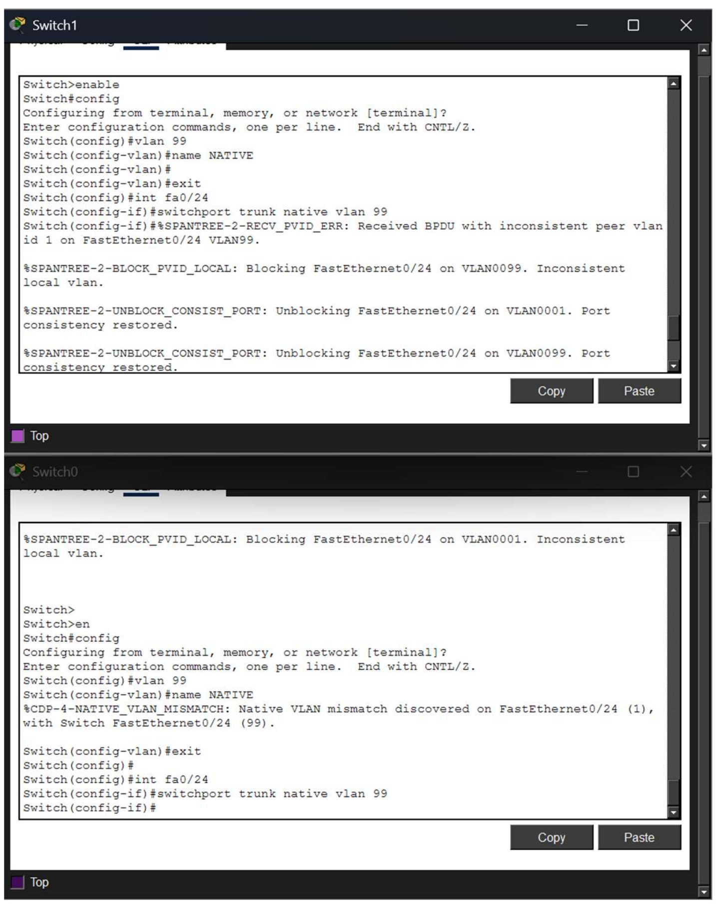
Configure Router to Switch Port as Trunk Port:

```
Switch>
Switch>en
Switch#config
Configuring from terminal, memory, or network [terminal]?
Enter configuration commands, one per line.  End with CNTL/Z.
Switch(config)#int fa0/23
Switch(config-if)#switchport mode trunk
Switch(config-if)#exit
Switch(config)#exit
Switch#
%SYS-5-CONFIG_I: Configured from console by console
```

Ping test:



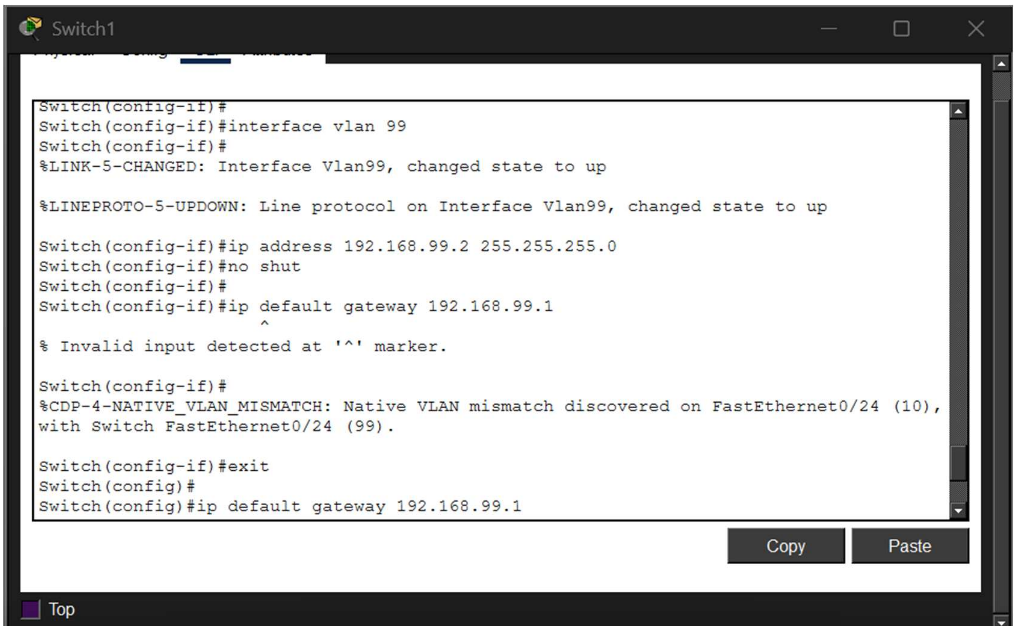
Changing the Native VLAN port to 99 from 10:

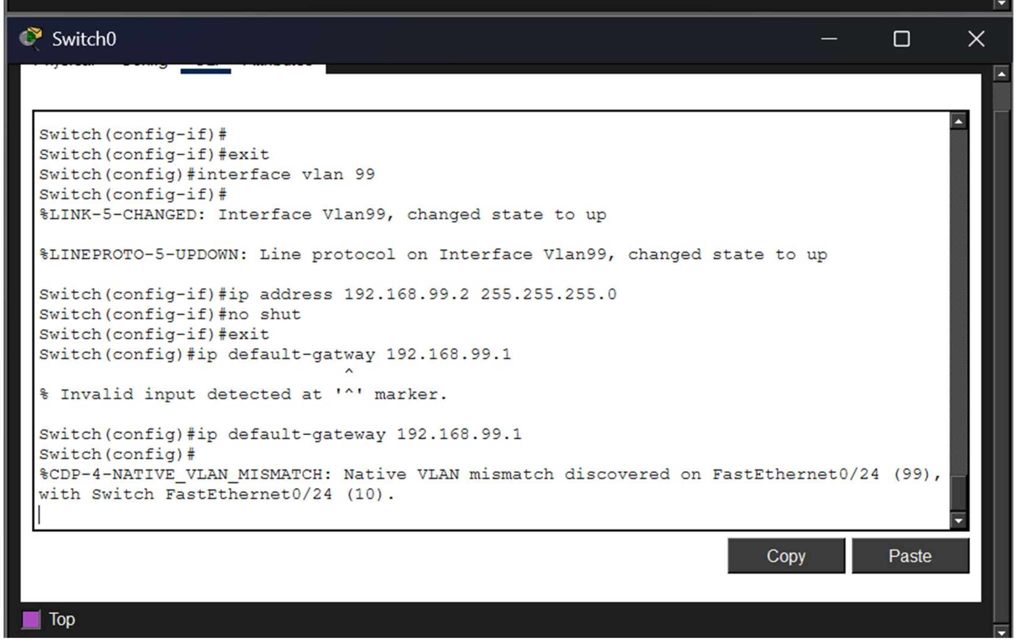


Again, Changing in One switch alone to 10:

```
Switch(config-if)#exit
Switch(config)#
Switch(config)#interface fa0/24
Switch(config-if)#switchport trunk native vlan 10
Switch(config-if)#%SPANTREE-2-RECV_PVID_ERR: Received BPDU with inconsistent peer vlan
id 99 on FastEthernet0/24 VLAN10.

%SPANTREE-2-BLOCK_PVID_LOCAL: Blocking FastEthernet0/24 on VLAN0010. Inconsistent
local vlan.


%CDP-4-NATIVE_VLAN_MISMATCH: Native VLAN mismatch discovered on FastEthernet0/24 (10),
with Switch FastEthernet0/24 (99).
```

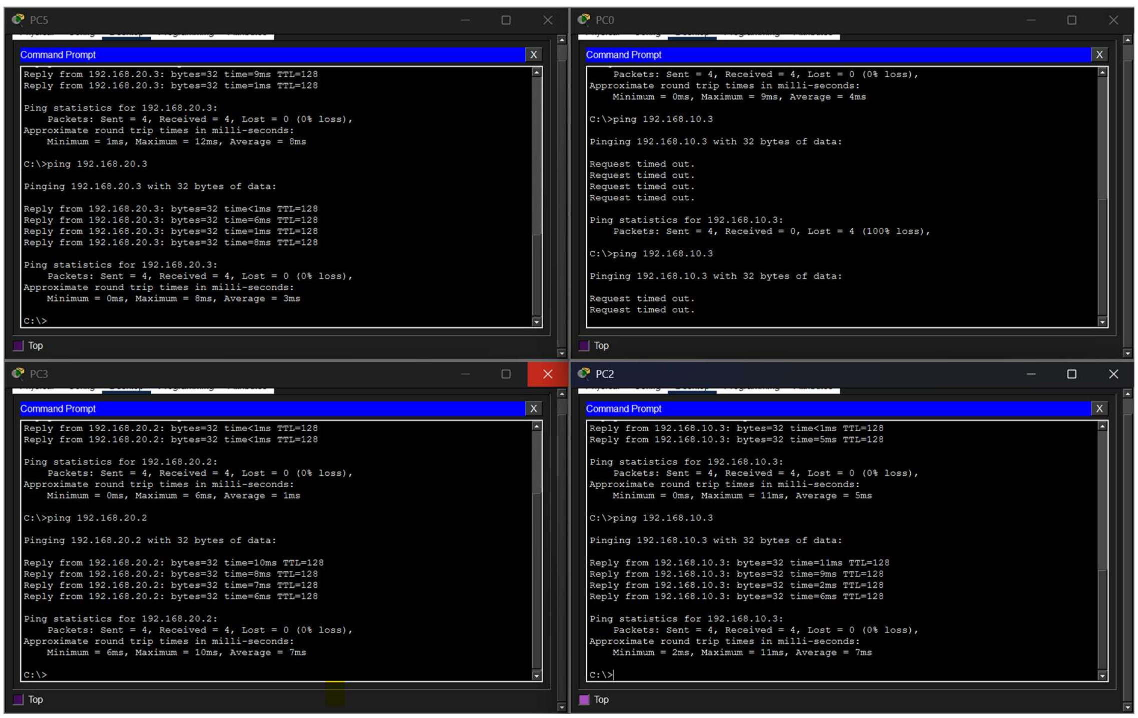
Hence, Troubleshoot:





```
Router>
Router>en
Router#config
Configuring from terminal, memory, or network [terminal]?
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#int g0/0.99
Router(config-subif)#
%LINK-3-UPDOWN: Interface GigabitEthernet0/0.99, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0.99, changed state to
up

Router(config-subif)#encapsulation dot1Q 99 native
Router(config-subif)#ip address 192.168.99.1 255.255.255.0
Router(config-subif)#
Router(config-subif)#exit
Router(config)#exit
Router#
%SYS-5-CONFIG_I: Configured from console by console

Router#
Router#
```

Ping Test Once again:



Configure SSH/Telnet:

Switch - 1

Switch1

Physical Config CLI Attributes

```
switch1#
switch1#en
switch1#config
Configuring from terminal, memory, or network [terminal]?
Enter configuration commands, one per line.  End with CNTL/Z.
switch1(config)#hostname switch1
switch1(config)#ip domain-name lab.local
switch1(config)#crypto key generate rsa
% You already have RSA keys defined named switch1.lab.local .
% Do you really want to replace them? [yes/no]: y
The name for the keys will be: switch1.lab.local
Choose the size of the key modulus in the range of 360 to 4096 for your
  General Purpose Keys. Choosing a key modulus greater than 512 may take
  a few minutes.

How many bits in the modulus [512]: 1024
% Generating 1024 bit RSA keys, keys will be non-exportable...[OK]

switch1(config)#
%CDP-4-NATIVE_VLAN_MISMATCH: Native VLAN mismatch discovered on FastEthernet0/24 (10), with Switch FastEthernet0/24 (99).

*Mar 1 1:25:2.368: %SSH-5-ENABLED: SSH 1.99 has been enabled
switch1(config)#username admin password 1234
switch1(config)#lint vty 0 4
                   ^
% Invalid input detected at '^' marker.

switch1(config)#line vty 0 4
switch1(config-line)#login local
switch1(config-line)#transport input telnet ssh
                                            ^
% Invalid input detected at '^' marker.

switch1(config-line)#
%CDP-4-NATIVE_VLAN_MISMATCH: Native VLAN mismatch discovered on FastEthernet0/24 (10), with Switch FastEthernet0/24 (99).

switch1(config-line)#
switch1(config-line)#transport input all
switch1(config-line)#
%CDP-4-NATIVE_VLAN_MISMATCH: Native VLAN mismatch discovered on FastEthernet0/24 (10), with Switch FastEthernet0/24 (99).
```

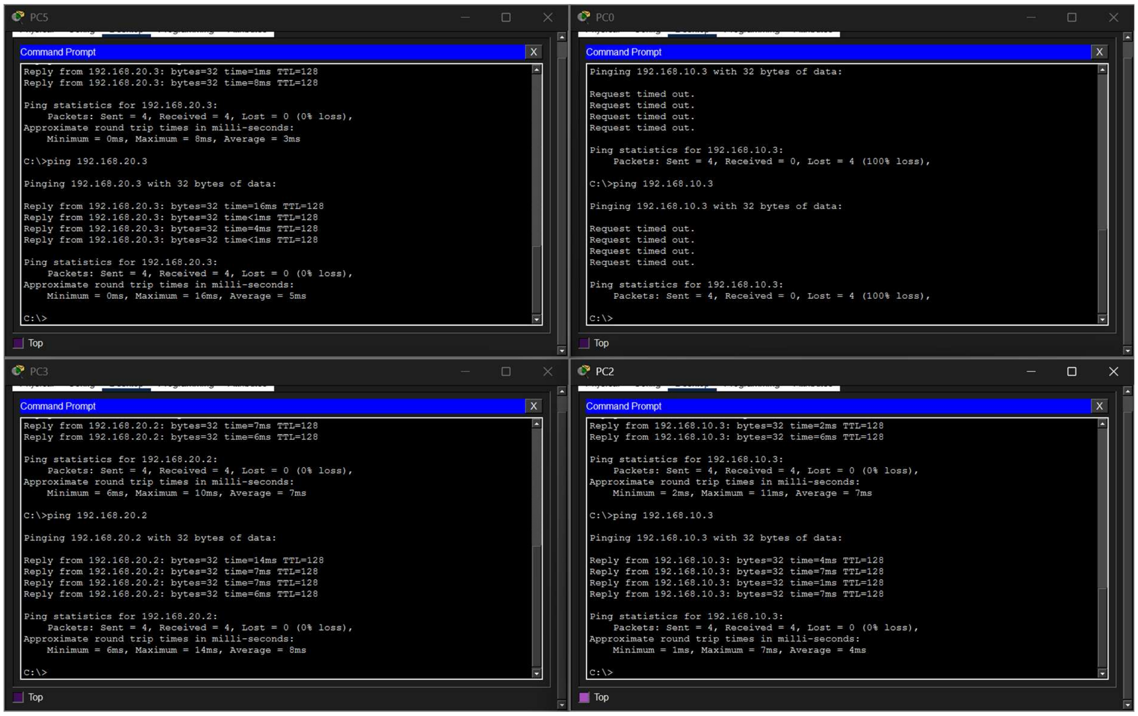
Switch – 2:

Switch0

Physical Config CLI Attributes

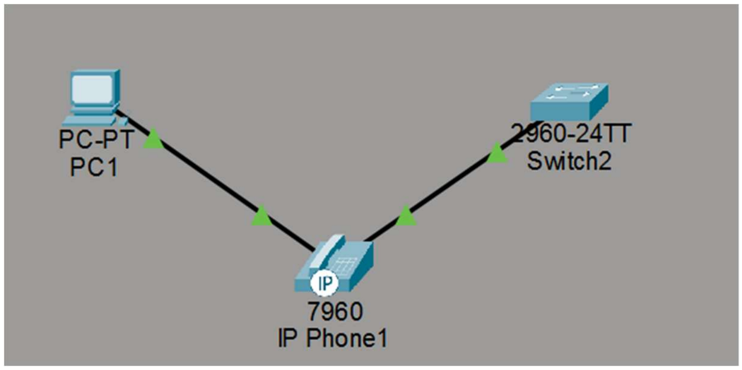
```
*Mar 1 1:27:43.874: %SSH-5-ENABLED: SSH 1.99 has been enabled
switch1(config)#exit
switch1#
%SYS-5-CONFIG_I: Configured from console by console

switch1#line vty 0 4
             ^
% Invalid input detected at '^' marker.

switch1#en
switch1#config
Configuring from terminal, memory, or network [terminal]?
Enter configuration commands, one per line.  End with CNTL/Z.
switch1(config)#hostname switch1
switch1(config)#username admin password 1234
switch1(config)#exit
switch1#
%SYS-5-CONFIG_I: Configured from console by console

switch1#en
switch1#con
%CDP-4-NATIVE_VLAN_MISMATCH: Native VLAN mismatch discovered on FastEthernet0/24 (99), with switch1 FastEthernet0/24 (10).
fig
Configuring from terminal, memory, or network [terminal]?
Enter configuration commands, one per line.  End with CNTL/Z.
switch1(config)#hostname Switch1
Switch1(config)#ip domain-name lab.local
Switch1(config)#username admin password 1234
Switch1(config)#crypto key generate rsa
% You already have RSA keys defined named switch1.lab.local .
% Do you really want to replace them? [yes/no]: y
The name for the keys will be: Switch1.lab.local
Choose the size of the key modulus in the range of 360 to 4096 for your
  General Purpose Keys. Choosing a key modulus greater than 512 may take
  a few minutes.

How many bits in the modulus [512]: 1024
%CDP-4-NATIVE_VLAN_MISMATCH: Native VLAN mismatch discovered on FastEthernet0/24 (99), with switch1 FastEthernet0/24 (10).

% Generating 1024 bit RSA keys, keys will be non-exportable...[OK]

Switch1(config)#line vty 0 4
*Mar 1 1:30:25.365: %SSH-5-ENABLED: SSH 1.99 has been enabled
Switch1(config-line)#transport input all
Switch1(config-line)#
%CDP-4-NATIVE_VLAN_MISMATCH: Native VLAN mismatch discovered on FastEthernet0/24 (99), with switch1 FastEthernet0/24 (10).
```

Ping Test:

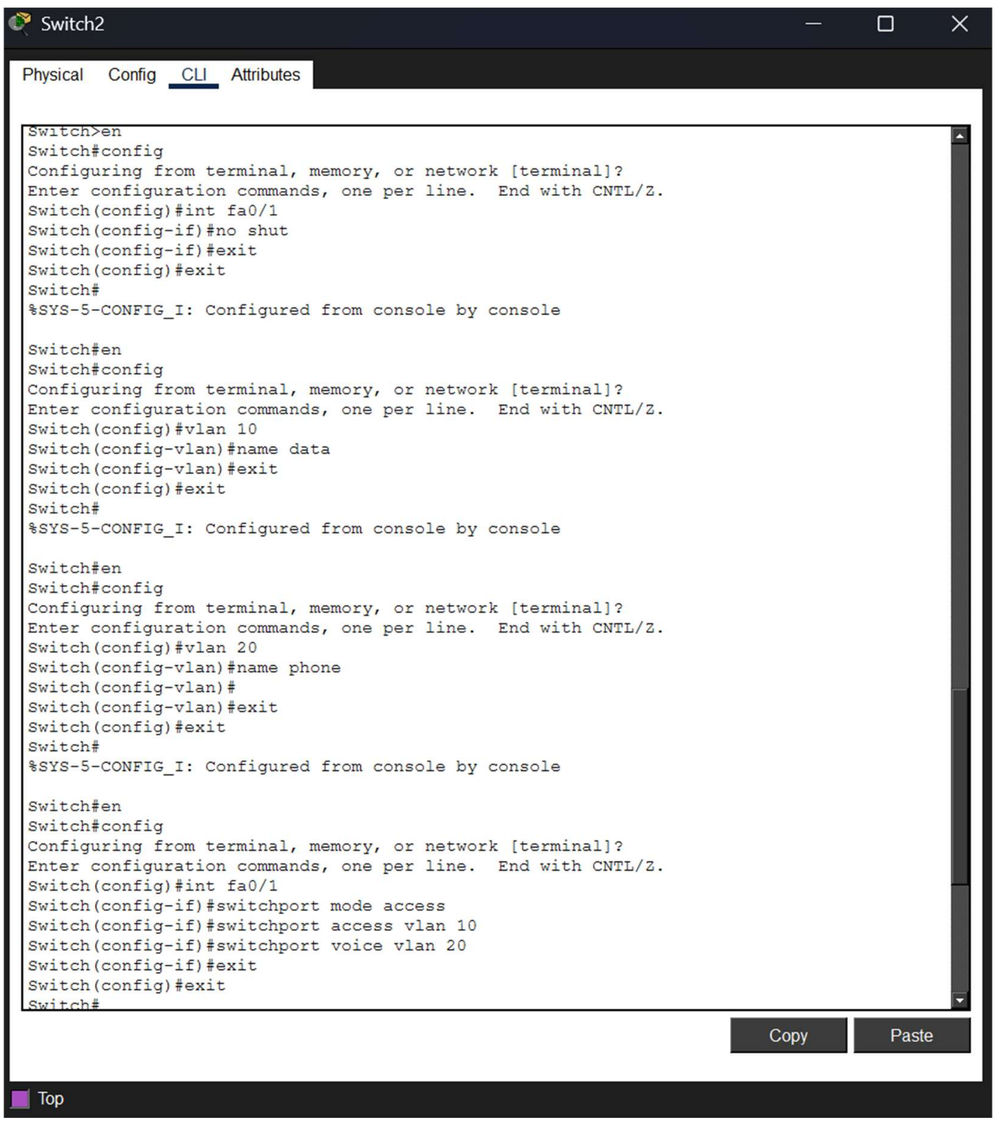

PC5
Command Prompt
Reply from 192.168.20.3: bytes=32 time=1ms TTL=128
Reply from 192.168.20.3: bytes=32 time=8ms TTL=128
Ping statistics for 192.168.20.3:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 0ms, Maximum = 8ms, Average = 3ms
C:\>ping 192.168.20.3
Pinging 192.168.20.3 with 32 bytes of data:
Reply from 192.168.20.3: bytes=32 time=16ms TTL=128
Reply from 192.168.20.3: bytes=32 time<1ms TTL=128
Reply from 192.168.20.3: bytes=32 time=4ms TTL=128
Reply from 192.168.20.3: bytes=32 time<1ms TTL=128
Ping statistics for 192.168.20.3:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 0ms, Maximum = 16ms, Average = 5ms
C:\>
Top
PC0
Command Prompt
Pinging 192.168.10.3 with 32 bytes of data:
Request timed out.
Request timed out.
Request timed out.
Request timed out.
Ping statistics for 192.168.10.3:
Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),
C:\>ping 192.168.10.3
Pinging 192.168.10.3 with 32 bytes of data:
Request timed out.
Request timed out.
Request timed out.
Request timed out.
Ping statistics for 192.168.10.3:
Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),
C:\>
Top
PC3
Command Prompt
Reply from 192.168.20.2: bytes=32 time=7ms TTL=128
Reply from 192.168.20.2: bytes=32 time=6ms TTL=128
Ping statistics for 192.168.20.2:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 6ms, Maximum = 10ms, Average = 7ms
C:\>ping 192.168.20.2
Pinging 192.168.20.2 with 32 bytes of data:
Reply from 192.168.20.2: bytes=32 time=14ms TTL=128
Reply from 192.168.20.2: bytes=32 time=7ms TTL=128
Reply from 192.168.20.2: bytes=32 time=7ms TTL=128
Reply from 192.168.20.2: bytes=32 time=6ms TTL=128
Ping statistics for 192.168.20.2:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 6ms, Maximum = 14ms, Average = 8ms
C:\>
Top
PC2
Command Prompt
Reply from 192.168.10.3: bytes=32 time=2ms TTL=128
Reply from 192.168.10.3: bytes=32 time=6ms TTL=128
Ping statistics for 192.168.10.3:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 2ms, Maximum = 11ms, Average = 7ms
C:\>ping 192.168.10.3
Pinging 192.168.10.3 with 32 bytes of data:
Reply from 192.168.10.3: bytes=32 time=4ms TTL=128
Reply from 192.168.10.3: bytes=32 time=7ms TTL=128
Reply from 192.168.10.3: bytes=32 time=1ms TTL=128
Reply from 192.168.10.3: bytes=32 time=7ms TTL=128
Ping statistics for 192.168.10.3:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 1ms, Maximum = 7ms, Average = 4ms
C:\>
Top

5. You have a Cisco switch and a VoIP phone that needs to be placed in a voice VLAN (VLAN 20). The data for the PC should remain in a separate VLAN (VLAN 10). Configure the switch port to support both voice and data traffic.

Network Topology:

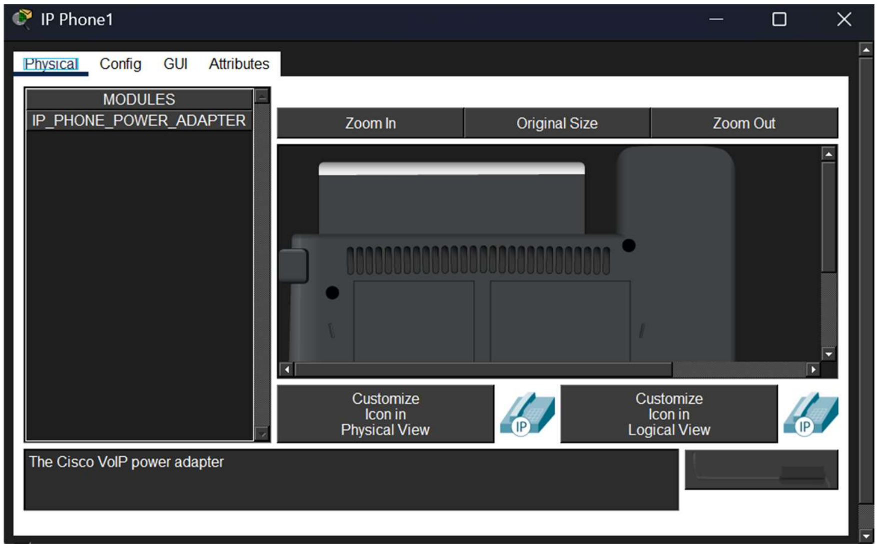


Switch Config:



```
Switch>en
Switch#config
Configuring from terminal, memory, or network [terminal]?
Enter configuration commands, one per line.  End with CNTL/Z.
Switch(config)#int fa0/1
Switch(config-if)#no shut
Switch(config-if)#exit
Switch(config)#exit
Switch#
%SYS-5-CONFIG_I: Configured from console by console

Switch#en
Switch#config
Configuring from terminal, memory, or network [terminal]?
Enter configuration commands, one per line.  End with CNTL/Z.
Switch(config)#vlan 10
Switch(config-vlan)#name data
Switch(config-vlan)#exit
Switch(config)#exit
Switch#
%SYS-5-CONFIG_I: Configured from console by console

Switch#en
Switch#config
Configuring from terminal, memory, or network [terminal]?
Enter configuration commands, one per line.  End with CNTL/Z.
Switch(config)#vlan 20
Switch(config-vlan)#name phone
Switch(config-vlan)#
Switch(config-vlan)#exit
Switch(config)#exit
Switch#
%SYS-5-CONFIG_I: Configured from console by console

Switch#en
Switch#config
Configuring from terminal, memory, or network [terminal]?
Enter configuration commands, one per line.  End with CNTL/Z.
Switch(config)#int fa0/1
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 10
Switch(config-if)#switchport voice vlan 20
Switch(config-if)#exit
Switch(config)#exit
Switch#
```

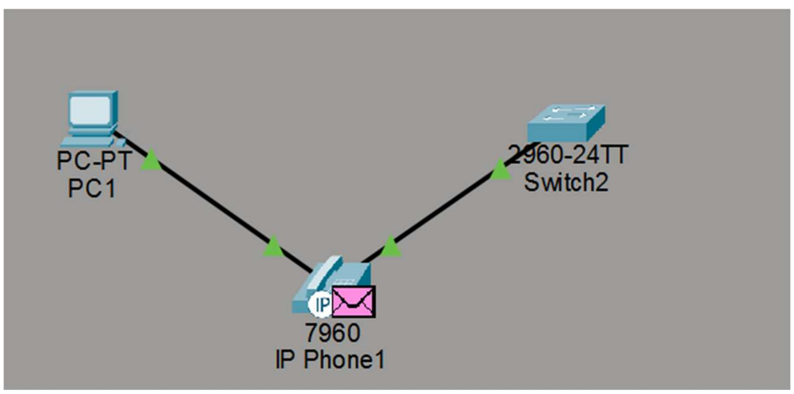
Hence, the adapter to the phone gets created:



Now, connecting the adapter to the phone:

Hence, the voip phone is powered on and works fine.

Working:



6. You configured VLANs 10 and 20 on your switch and assigned ports to each VLAN. However, devices in VLAN 10 cannot communicate with devices in VLAN 20. Troubleshoot the issue.

In order to enable inter vlan communication, we need a layer 3 device like router and a trunk link configured in it.

Step – 1: Check If VLAN exists with the command: 'sh vlan brief'

Step – 2: Check IP Configuration

- Devices in VLAN 10 → same subnet
- Devices in VLAN 20 → different subnet

Example:

VLAN	IP Range
10	192.168.10.0/24
20	192.168.20.0/24

Step – 4: Bring Router in:

```
interface fa0/0.10
encapsulation dot1Q 10
ip address 192.168.10.1 255.255.255.0

interface fa0/0.20
encapsulation dot1Q 20
ip address 192.168.20.1 255.255.255.0
```

Step 6: Check Trunk Link

show interfaces trunk

Ensure VLAN 10 & 20 allowed

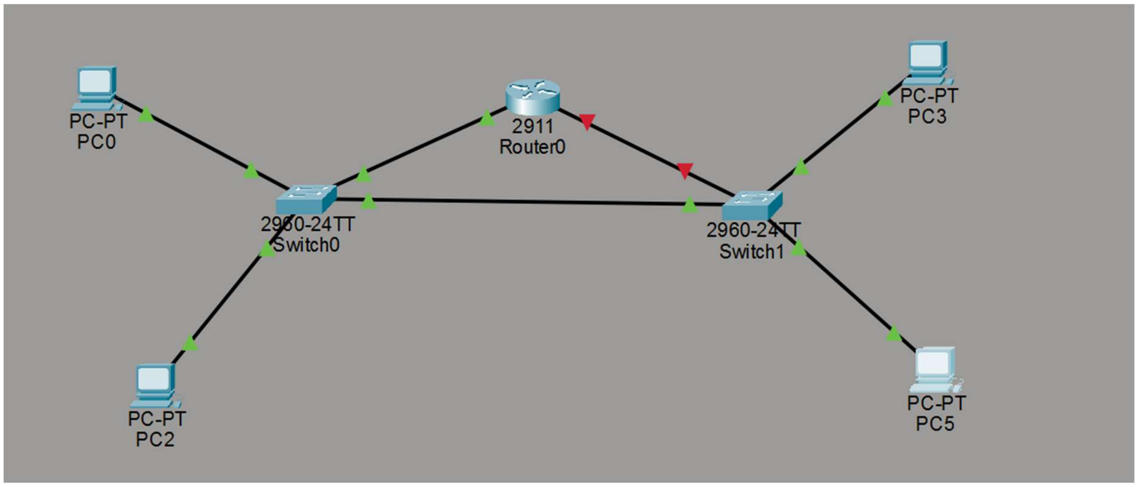
Step 7: Verify Default Gateway

Each device must have:

VLAN	Gateway
10	192.168.10.1
20	192.168.20.1

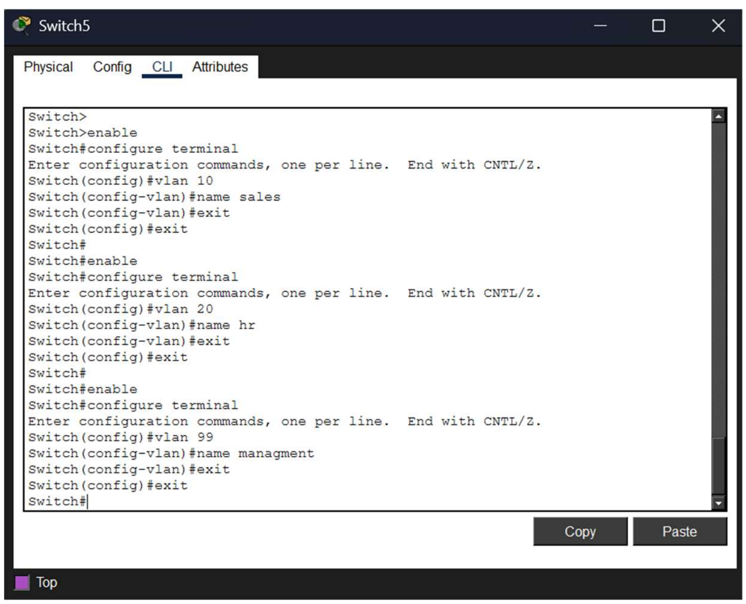
7. Try Inter VLAN routing with Router

Network Topology:



Switch - 1 Config:

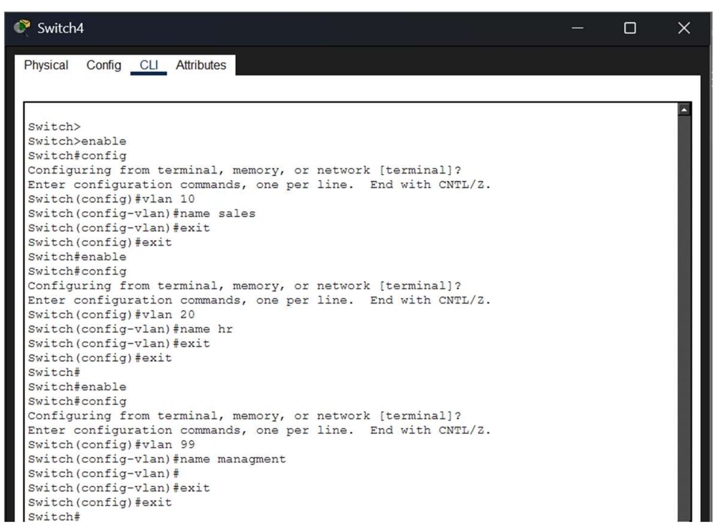
Set VLAN Identifier Names:



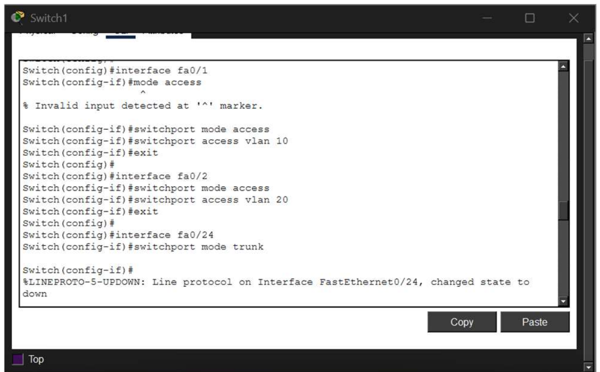
Set Port Types for every VLANs:

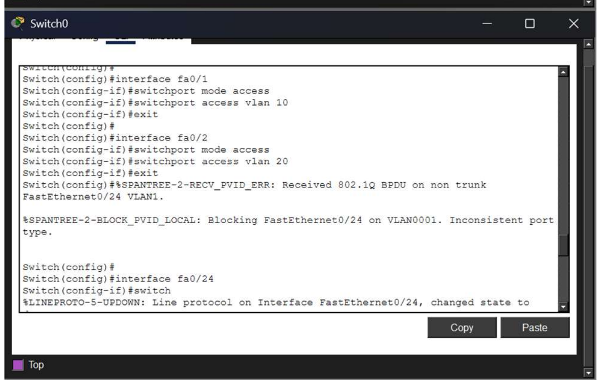
Switch -2 Config:

Create VLAN Identifiers:

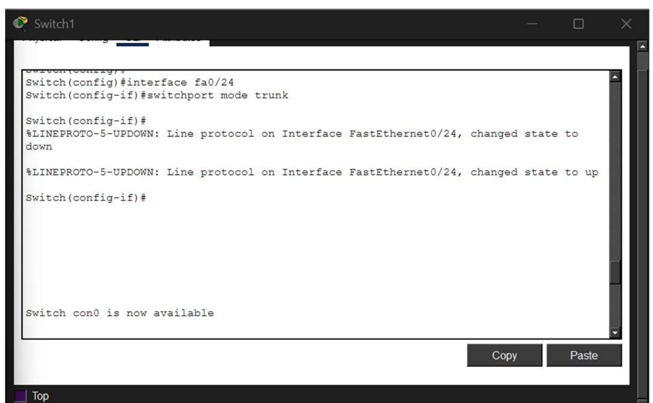


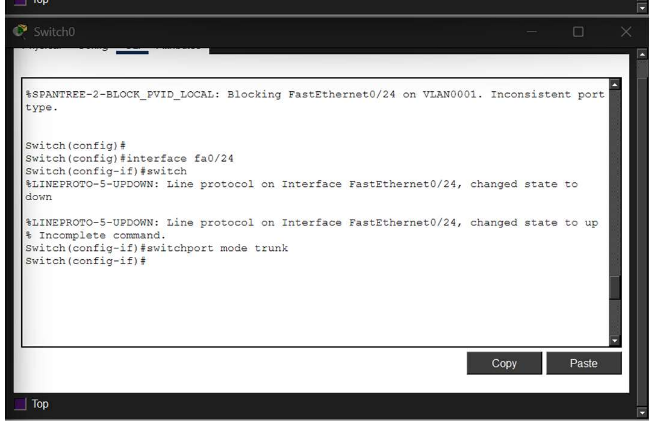
Assign Access Ports:



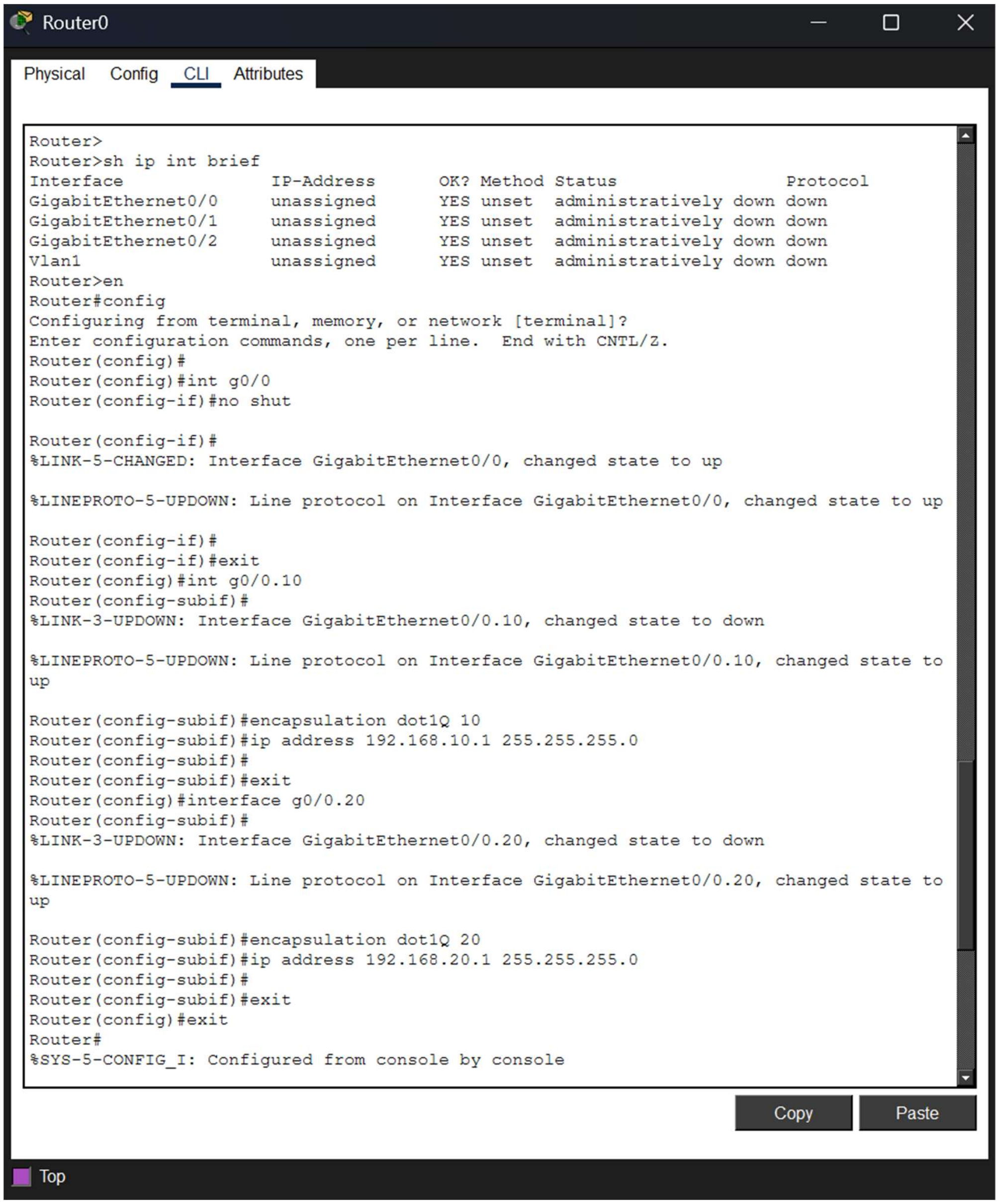


Assign Trunk Ports:





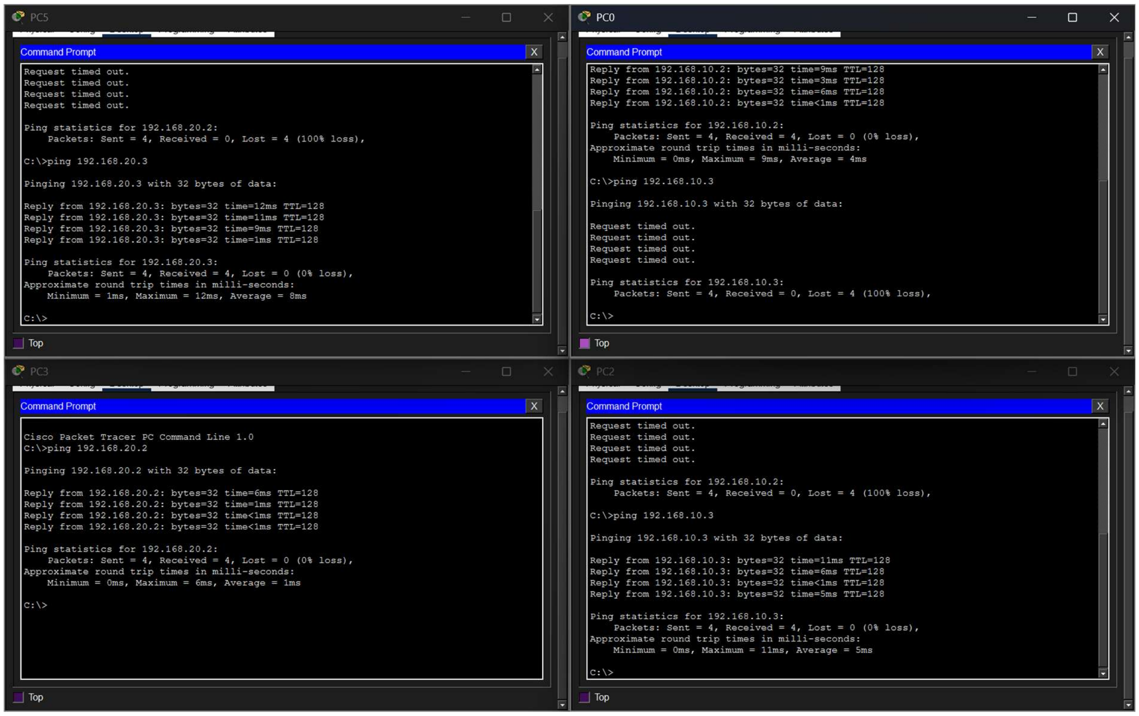
Router Config:



Configure Router to Switch Port as Trunk Port:

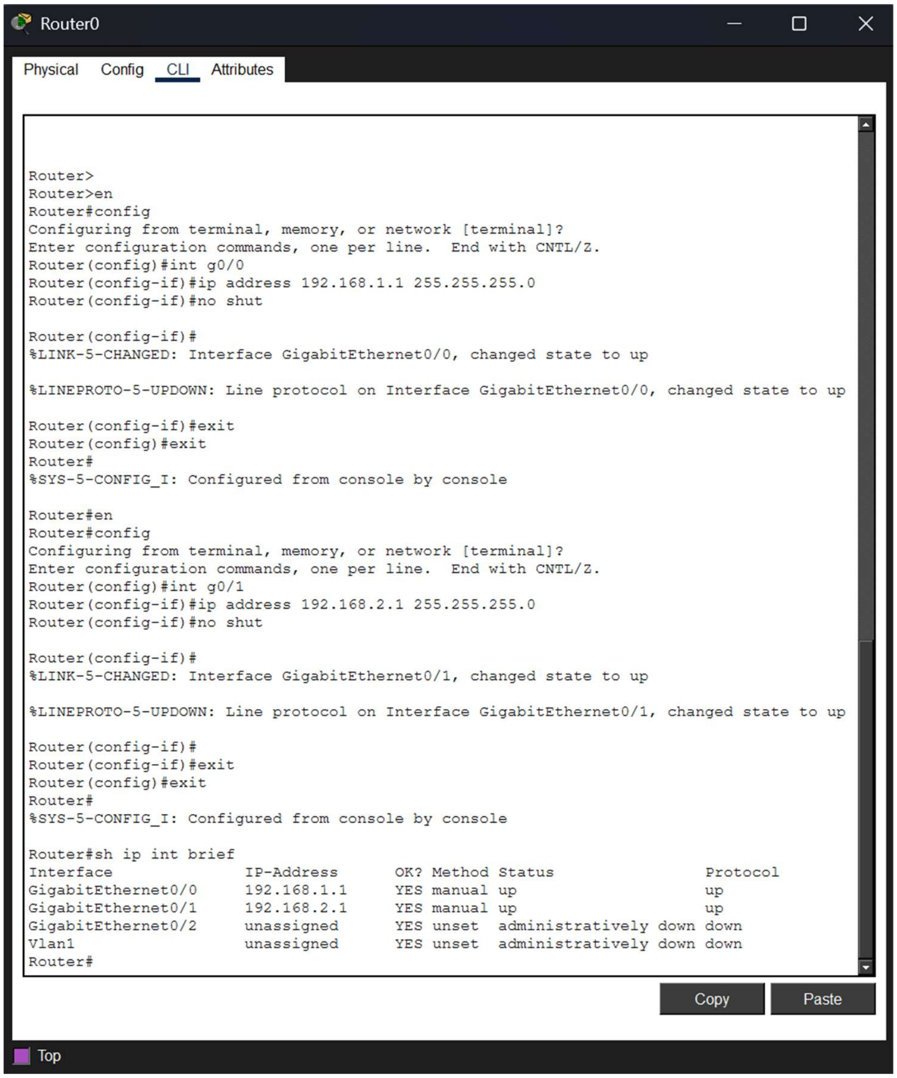
```
Switch>
Switch>en
Switch#config
Configuring from terminal, memory, or network [terminal]?
Enter configuration commands, one per line.  End with CNTL/Z.
Switch(config)#int fa0/23
Switch(config-if)#switchport mode trunk
Switch(config-if)#exit
Switch(config)#exit
Switch#
%SYS-5-CONFIG_I: Configured from console by console
```

Ping test:


PC5
Command Prompt
Request timed out.
Request timed out.
Request timed out.
Request timed out.
Ping statistics for 192.168.20.2:
Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),
C:\>ping 192.168.20.3
Pinging 192.168.20.3 with 32 bytes of data:
Reply from 192.168.20.3: bytes=32 time=12ms TTL=128
Reply from 192.168.20.3: bytes=32 time=11ms TTL=128
Reply from 192.168.20.3: bytes=32 time=9ms TTL=128
Reply from 192.168.20.3: bytes=32 time=1ms TTL=128
Ping statistics for 192.168.20.3:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 1ms, Maximum = 12ms, Average = 8ms
C:\>
Top
PC0
Command Prompt
Reply from 192.168.10.2: bytes=32 time=9ms TTL=128
Reply from 192.168.10.2: bytes=32 time=3ms TTL=128
Reply from 192.168.10.2: bytes=32 time=6ms TTL=128
Reply from 192.168.10.2: bytes=32 time<1ms TTL=128
Ping statistics for 192.168.10.2:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 0ms, Maximum = 9ms, Average = 4ms
C:\>ping 192.168.10.3
Pinging 192.168.10.3 with 32 bytes of data:
Request timed out.
Request timed out.
Request timed out.
Request timed out.
Ping statistics for 192.168.10.3:
Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),
C:\>
Top
PC3
Command Prompt
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.20.2
Pinging 192.168.20.2 with 32 bytes of data:
Reply from 192.168.20.2: bytes=32 time=6ms TTL=128
Reply from 192.168.20.2: bytes=32 time=1ms TTL=128
Reply from 192.168.20.2: bytes=32 time<1ms TTL=128
Reply from 192.168.20.2: bytes=32 time<1ms TTL=128
Ping statistics for 192.168.20.2:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 0ms, Maximum = 6ms, Average = 1ms
C:\>
Top
PC2
Command Prompt
Request timed out.
Request timed out.
Request timed out.
Request timed out.
Ping statistics for 192.168.10.2:
Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),
C:\>ping 192.168.10.3
Pinging 192.168.10.3 with 32 bytes of data:
Reply from 192.168.10.3: bytes=32 time=11ms TTL=128
Reply from 192.168.10.3: bytes=32 time=6ms TTL=128
Reply from 192.168.10.3: bytes=32 time<1ms TTL=128
Reply from 192.168.10.3: bytes=32 time=5ms TTL=128
Ping statistics for 192.168.10.3:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 0ms, Maximum = 11ms, Average = 5ms
C:\>
Top

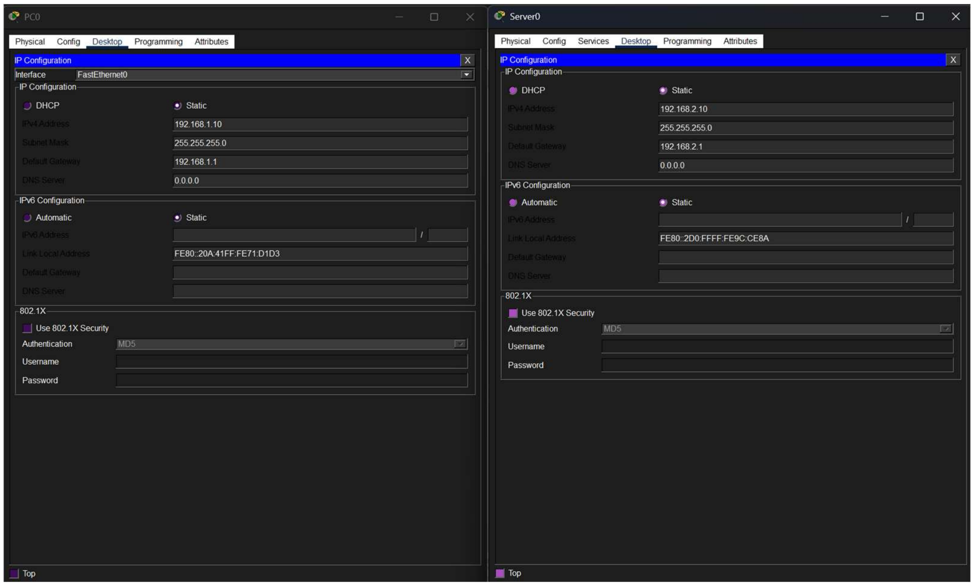
8. Implement ACLs to restrict traffic based on source and destination ports. Test rules by simulating legitimate and unauthorized traffic.

Router Config:




```
Router>
Router>en
Router#config
Configuring from terminal, memory, or network [terminal]?
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#int g0/0
Router(config-if)#ip address 192.168.1.1 255.255.255.0
Router(config-if)#no shut

Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up

Router(config-if)#exit
Router(config)#exit
Router#
%SYS-5-CONFIG_I: Configured from console by console

Router#en
Router#config
Configuring from terminal, memory, or network [terminal]?
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#int g0/1
Router(config-if)#ip address 192.168.2.1 255.255.255.0
Router(config-if)#no shut

Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/1, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to up

Router(config-if)#
Router(config-if)#exit
Router(config)#exit
Router#
%SYS-5-CONFIG_I: Configured from console by console

Router#sh ip int brief
Interface              IP-Address      OK? Method Status                Protocol
GigabitEthernet0/0     192.168.1.1     YES manual up                    up
GigabitEthernet0/1     192.168.2.1     YES manual up                    up
GigabitEthernet0/2     unassigned      YES unset  administratively down down
Vlan1                  unassigned      YES unset  administratively down down
Router#
```

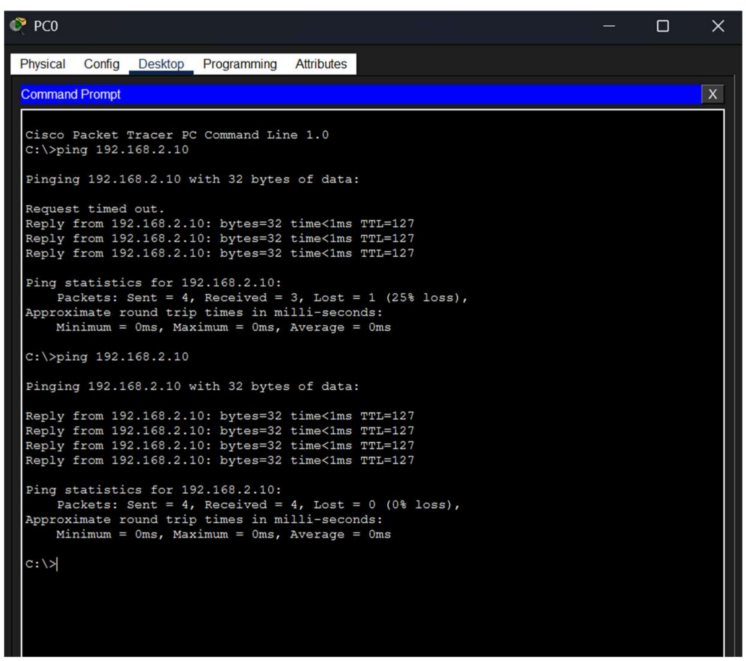
PC Configs:



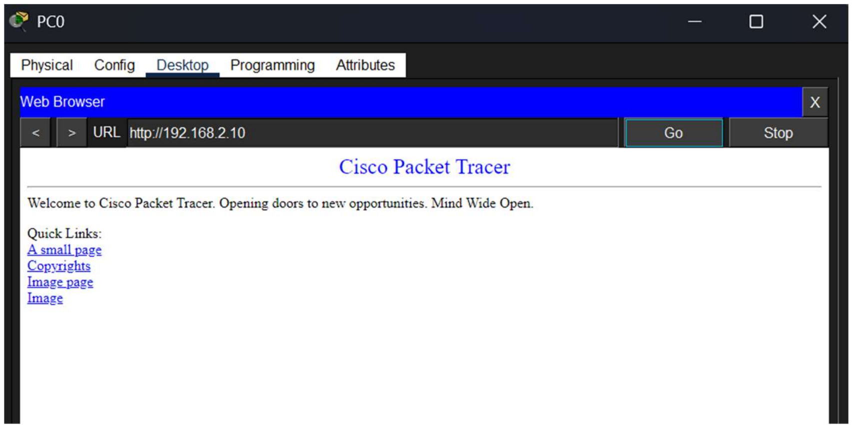
Configure and Enable ACL:

```
Router#en
Router#config
Configuring from terminal, memory, or network [terminal]?
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#access-list 100 permit tcp 192.168.1.0 0.0.0.255 host 192.168.2.10 eq 80
Router(config)#access-list 100 deny tcp 192.168.1.0 0.0.0.255 host 192.168.2.10 eq 23
Router(config)#access-list 100 permit ip any any
Router(config)#
Router(config)#int g0/0
Router(config-if)#ip access-group 100 in
Router(config-if)#
```

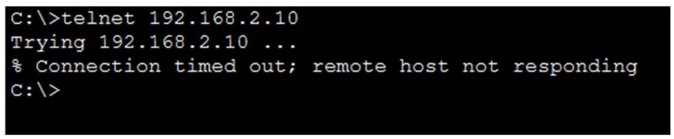
Ping Test from PC to Server:



Checking if the Web Server is Reachable:

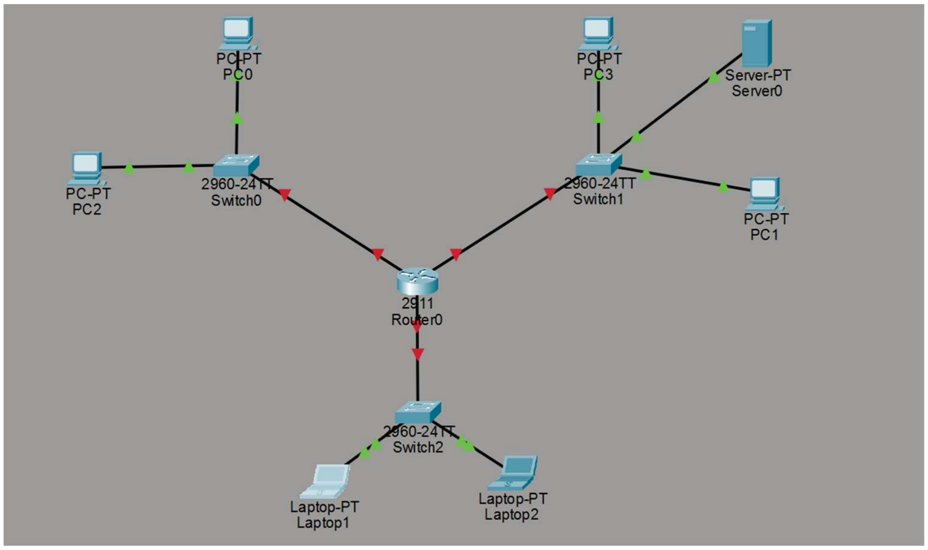


Testing Telnet and Successfully Blocked:



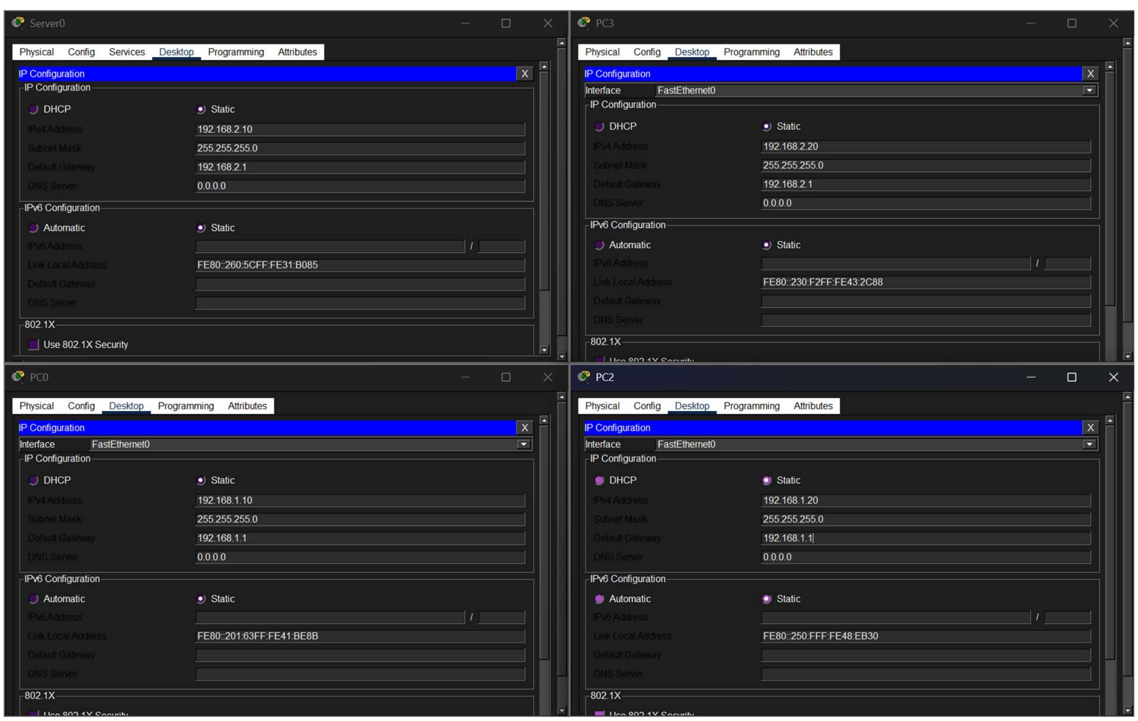
9. Configure a standard Access Control List (ACL) on a router to permit traffic from a specific IP range.Test connectivity to verify the ACL is working as intended.

Network Topology:

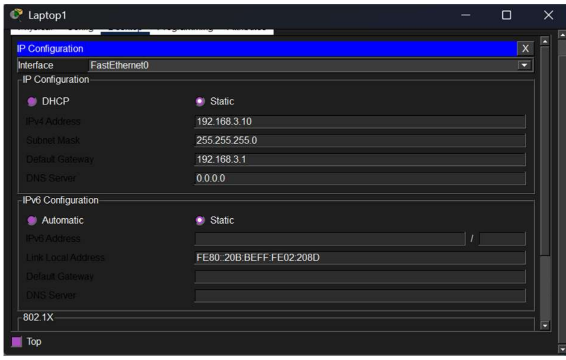


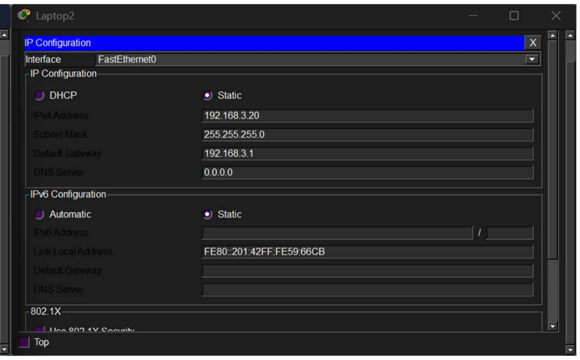
Legal: 192.168.1.0/24, 192.168.20/24 and Outsiders: 192.168.3.0/24

PC Config:



Network Config of Outsider Network PCs:





Router Config:

Routing Table:

```
Router#
Router#sh ip int brief
Interface              IP-Address      OK? Method Status                Protocol
GigabitEthernet0/0     192.168.1.1     YES manual up                    up
GigabitEthernet0/1     192.168.2.1     YES manual up                    up
GigabitEthernet0/2     192.168.3.1     YES manual up                    up
Vlan1                  unassigned      YES unset  administratively down down
Router#
```

Creating ACL and Assigning to Interface:

```
Router#en
Router#config
Configuring from terminal, memory, or network [terminal]?
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#access-list 10 permit 192.168.1.0 0.0.0.255
Router(config)#access-list 10 deny any
Router(config)#
Router(config)#ing g0/1
                ^
% Invalid input detected at '^' marker.

Router(config)#int g0/1
Router(config-if)#ip access-group 10 out
Router(config-if)#exit
Router(config)#exit
Router#
%SYS-5-CONFIG_I: Configured from console by console

Router#
```

Ping Test from 192.168.1.0/24 network PCs:

```
C:\>ping 192.168.2.10

Pinging 192.168.2.10 with 32 bytes of data:

Request timed out.
Reply from 192.168.2.10: bytes=32 time=8ms TTL=127
Reply from 192.168.2.10: bytes=32 time=8ms TTL=127
Reply from 192.168.2.10: bytes=32 time=8ms TTL=127

Ping statistics for 192.168.2.10:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
Approximate round trip times in milli-seconds:
    Minimum = 8ms, Maximum = 8ms, Average = 8ms
```

Ping from 192.168.3.0/24 network PCs:

10. Create an extended ACL to block specific applications, such as HTTP or FTP traffic. Test the ACL rules by attempting to access blocked services.

Router Config:

R1

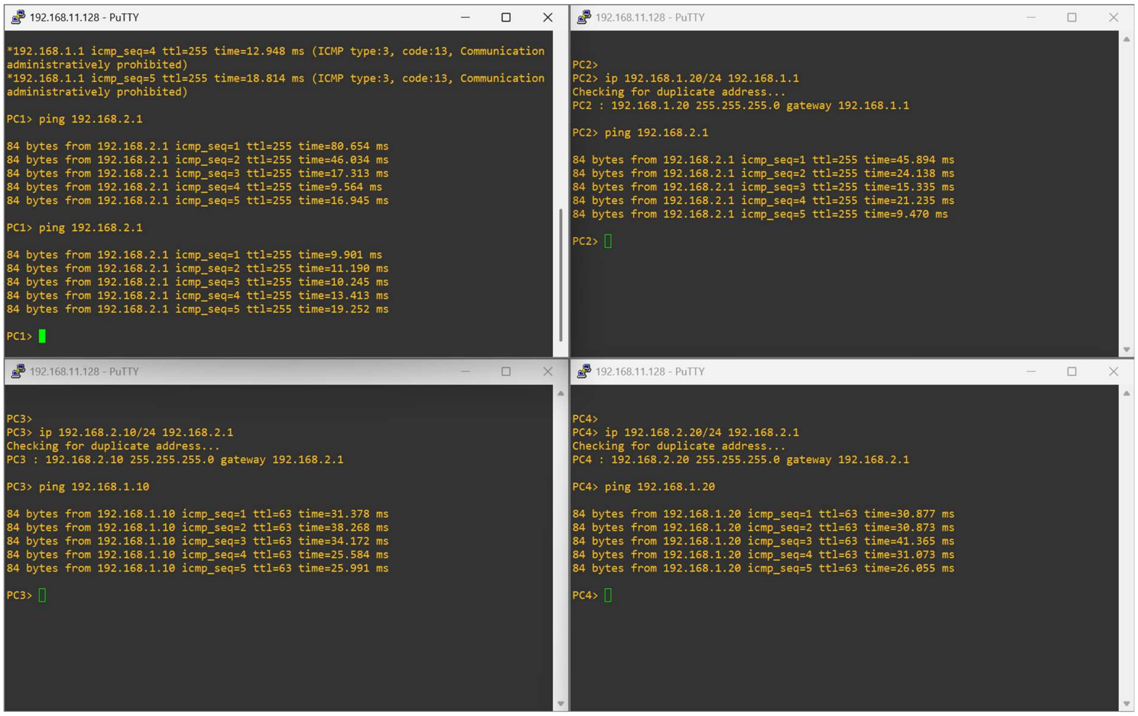
```
R1#config
Configuring from terminal, memory, or network [terminal]?
Enter configuration commands, one per line.  End with CNTL/Z.
R1(config)#int f0/0
R1(config-if)#ip address 192.168.1.1 255.255.255.0
R1(config-if)#no shut
R1(config-if)#exit
R1(config)#exig
               ^
% Invalid input detected at '^' marker.

R1(config)#
*Mar 27 18:20:35.047: %LINK-3-UPDOWN: Interface FastEthernet0/0, changed state t
o up
*Mar 27 18:20:36.047: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthern
et0/0, changed state to up
R1(config)#exit
R1#
R1#
*Mar 27 18:20:37.911: %SYS-5-CONFIG_I: Configured from console by console
R1#en
R1#config
Configuring from terminal, memory, or network [terminal]?
Enter configuration commands, one per line.  End with CNTL/Z.
R1(config)#int f0/1
R1(config-if)#ip address 192.168.2.1 255.255.255.0
R1(config-if)#no sht=ut
                    ^
% Invalid input detected at '^' marker.

R1(config-if)#no shut
R1(config-if)#exit
R1(config)#exit
R1#
*Mar 27 18:21:09.731: %SYS-5-CONFIG_I: Configured from console by console
R1#
R1#
*Mar 27 18:21:09.739: %LINK-3-UPDOWN: Interface FastEthernet0/1, changed state t
o up
*Mar 27 18:21:10.739: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthern
et0/1, changed state to up
R1#
```

```
R1#sh ip int brief
Interface              IP-Address      OK? Method Status                Prot
ocol
FastEthernet0/0        192.168.1.1     YES manual up                    up

FastEthernet0/1        192.168.2.1     YES manual up                    up

R1#
```

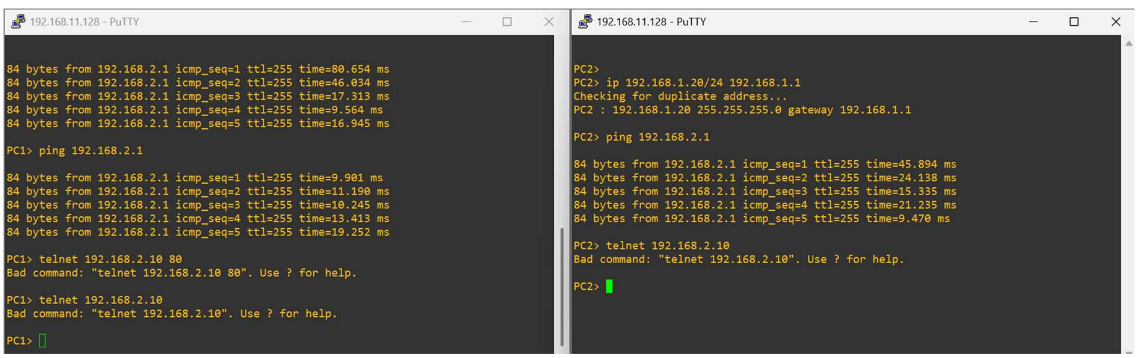
Config ACL to prevent HTTP and FTP

```
R1#en
R1#config
Configuring from terminal, memory, or network [terminal]?
Enter configuration commands, one per line.  End with CNTL/Z.
R1(config)#access-list 100 deny 192.168.1.0 0.0.0.255 host 192.168.2.10 eq 80
                                  ^
% Invalid input detected at '^' marker.

R1(config)#access-list 100 deny tcp  192.168.1.0 0.0.0.255 host 192.168.2.10 e$
R1(config)#$ 100 deny tcp  192.168.1.0 0.0.0.255 host 192.168.2.10 eq 21
R1(config)#$ 100 permit ip any any
R1(config)#
R1(config)#exit
R1#
*Mar 27 18:37:42.703: %SYS-5-CONFIG_I: Configured from console by console
R1#
```

Ping Test – 1 Ping within Local LANs:



Telnet Blocked:



11. Try Static NAT, Dynamic NAT and PAT to translate Ips

Router Config:

```
R1
R1#en
R1#config
Configuring from terminal, memory, or network [terminal]?
Enter configuration commands, one per line.  End with CNTL/Z.
R1(config)#int f0/0
R1(config-if)#ip address 192.168.1.1 255.255.255.0
R1(config-if)#no shut
R1(config-if)#ip nat inside
*Mar 27 17:34:46.315: %LINK-3-UPDOWN: Interface FastEthernet0/0, changed state t
o up
*Mar 27 17:34:47.315: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthern
et0/0, changed state to up
R1(config-if)#ip nat inside
R1(config-if)#
*Mar 27 17:34:48.839: %LINEPROTO-5-UPDOWN: Line protocol on Interface NVI0, chan
ged state to up
R1(config-if)#
R1(config-if)#exit
R1(config)#exit
R1#
R1#en
*Mar 27 17:34:55.891: %SYS-5-CONFIG_I: Configured from console by console
R1#en c
      ^
% Invalid input detected at '^' marker.

R1#en
R1#config
Configuring from terminal, memory, or network [terminal]?
Enter configuration commands, one per line.  End with CNTL/Z.
R1(config)#int f0/1
R1(config-if)#ip address 200.1.1.1 255.255.255.0
R1(config-if)#no shut
R1(config-if)#ip nat outsid
*Mar 27 17:35:29.823: %LINK-3-UPDOWN: Interface FastEthernet0/1, changed state t
o up
*Mar 27 17:35:30.823: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthern
et0/1, changed state to up
R1(config-if)#ip nat outside
R1(config-if)#
```

PC Configs:

```
192.168.11.128 - PuTTY
Source code and license can be found at vpcs.sf.net.
For more information, please visit wiki.freecode.com.cn.

Press '?' to get help.

Executing the startup file


PC1> ip 192.168.1.10/24 192.168.1.1
Checking for duplicate address...
PC1 : 192.168.1.10 255.255.255.0 gateway 192.168.1.1

PC1> show ip

NAME        : PC1[1]
IP/MASK     : 192.168.1.10/24
GATEWAY     : 192.168.1.1
DNS         :
MAC         : 00:50:79:66:68:00
LPORT       : 20002
RHOST:PORT  : 127.0.0.1:20003
MTU         : 1500

PC1>
```

```
192.168.11.128 - PuTTY
Source code and license can be found at vpcs.sf.net.
For more information, please visit wiki.freecode.com.cn.

Press '?' to get help.

Executing the startup file


PC2> ip 192.168.1.20/24 192.168.1.1
Checking for duplicate address...
PC2 : 192.168.1.20 255.255.255.0 gateway 192.168.1.1

PC2> show ip

NAME        : PC2[1]
IP/MASK     : 192.168.1.20/24
GATEWAY     : 192.168.1.1
DNS         :
MAC         : 00:50:79:66:68:01
LPORT       : 20004
RHOST:PORT  : 127.0.0.1:20005
MTU         : 1500

PC2>
```

Static NAT Config:

```
R1#en
R1#config
Configuring from terminal, memory, or network [terminal]?
Enter configuration commands, one per line.  End with CNTL/Z.
R1(config)#int f0/0
R1(config-if)#ip nat inside source static 192.168.1.10 200.1.1.100
R1(config)#
R1(config)#exit
R1#e
*Mar 27 17:39:47.263: %SYS-5-CONFIG_I: Configured from console by console
R1#show ip nat translations
Pro Inside global      Inside local       Outside local      Outside global
--- 200.1.1.100        192.168.1.10       ---                ---
R1#
```

Configure Dynamic NATs:

Step – 1: Create an ACL

Step – 2: Create a NAT pool consisting some number of Ips

Step – 3: Bind the ACL Pool

```
R1#en
R1#config
Configuring from terminal, memory, or network [terminal]?
Enter configuration commands, one per line.  End with CNTL/Z.
R1(config)#access-list 1 permit 192.168.1.0 0.0.0.255
R1(config)#ip nat pool pool1 200.1.1.100 200.1.1.150 netmask 255.255.255.0
R1(config)#
R1(config)#ip nat inside source list 1 pool pool1
R1(config)#
```

PAT:

```
R1#en
R1#config
Configuring from terminal, memory, or network [terminal]?
Enter configuration commands, one per line.  End with CNTL/Z.
R1(config)#access-list 1 permit 192.168.1.0 0.0.0.255
R1(config)#ip nat inside source list 1 interface f0/1 overload
R1(config)#exit
R1#exit
```

End Router Config Status:

```
R1#sh ip int brief
Interface              IP-Address      OK? Method Status                Protocol
FastEthernet0/0        192.168.1.1     YES manual up                    up
FastEthernet0/1        200.1.1.1       YES manual up                    up
NVI0                   192.168.1.1     YES unset  up                    up
R1#show nat translations
         ^
% Invalid input detected at '^' marker.

R1#show ip nat translations
Pro Inside global      Inside local       Outside local      Outside global
--- 200.1.1.100        192.168.1.10       ---                ---
R1#
```

Ping Test:

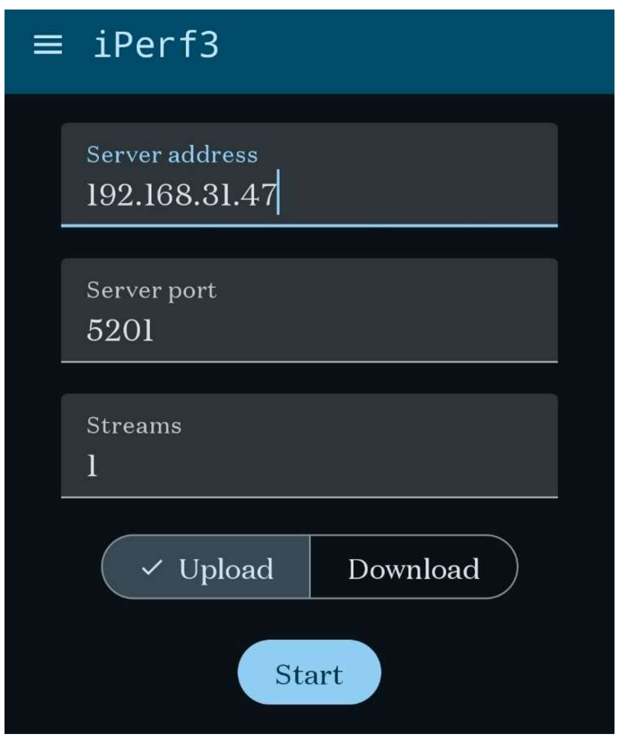
```
PC1> ping 200.1.1.1

84 bytes from 200.1.1.1 icmp_seq=1 ttl=255 time=20.031 ms
84 bytes from 200.1.1.1 icmp_seq=2 ttl=255 time=26.574 ms
84 bytes from 200.1.1.1 icmp_seq=3 ttl=255 time=21.838 ms
84 bytes from 200.1.1.1 icmp_seq=4 ttl=255 time=10.144 ms
84 bytes from 200.1.1.1 icmp_seq=5 ttl=255 time=25.481 ms
```

12. Download iperf in laptop/phone and make sure they are in same network. Try different iperf commands with tcp, udp, birectional, reverse, multicast, parallel

Start the Iperf Server:

```
kishore@kishore-virtual-machine:~$ ip addr
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
       valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host
       valid_lft forever preferred_lft forever
2: ens33: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 00:0c:29:71:b7:32 brd ff:ff:ff:ff:ff:ff
    altname enp2s1
    inet 192.168.31.47/24 brd 192.168.31.255 scope global dynamic noprefixroute ens33
       valid_lft 28433sec preferred_lft 28433sec
    inet6 2409:40f4:1127:484f:f12d:f167:8200:157c/64 scope global temporary dynamic
       valid_lft 16679sec preferred_lft 16679sec
    inet6 2409:40f4:1127:484f:12ac:61da:34da:357/64 scope global dynamic mngtmpaddr noprefixroute
       valid_lft 16679sec preferred_lft 16679sec
    inet6 fe80::fe95:eaf3:e714:9839/64 scope link noprefixroute
       valid_lft forever preferred_lft forever
kishore@kishore-virtual-machine:~$
kishore@kishore-virtual-machine:~$ iperf3 -s
-----------------------------------------------------------
Server listening on 5201
-----------------------------------------------------------
Accepted connection from 192.168.31.50, port 37920
[  5] local 192.168.31.47 port 5201 connected to 192.168.31.50 port 37922
[ ID] Interval           Transfer     Bitrate
[  5]   0.00-1.03   sec  4.62 MBytes  37.8 Mbits/sec
[  5]   1.03-2.03   sec  7.75 MBytes  64.9 Mbits/sec
[  5]   2.03-3.00   sec  7.62 MBytes  65.7 Mbits/sec
[  5]   3.00-4.02   sec  9.50 MBytes  78.6 Mbits/sec
[  5]   4.02-5.03   sec  5.00 MBytes  41.2 Mbits/sec
[  5]   5.03-6.02   sec  5.88 MBytes  50.1 Mbits/sec
[  5]   6.02-7.03   sec  6.00 MBytes  49.8 Mbits/sec
[  5]   7.03-8.05   sec  7.38 MBytes  60.8 Mbits/sec
[  5]   8.05-9.11   sec  6.62 MBytes  52.5 Mbits/sec
[  5]   9.11-10.03  sec  5.50 MBytes  50.0 Mbits/sec
[  5]  10.03-10.09  sec   384 KBytes  50.4 Mbits/sec
- - - - - - - - - - - - - - - - - - - - - - - - -
[ ID] Interval           Transfer     Bitrate
[  5]   0.00-10.09  sec  66.2 MBytes  55.1 Mbits/sec                  receiver
-----------------------------------------------------------
Server listening on 5201
-----------------------------------------------------------
```

Client Control:




Result
52.3 Mbps
Avg: 56.2
Min: 39.8
Max: 81.8
Interval
Transfer
Bitrate
0.00-1.00s 5.24 MBytes
1.00-2.00s 8.51 MBytes
2.00-3.00s 8.25 MBytes
3.00-4.00s 10.2 MBytes
4.00-5.00s 4.98 MBytes
5.00-6.00s 6.29 MBytes
6.00-7.00s 6.16 MBytes
7.00-8.00s 7.47 MBytes
8.00-9.00s 6.68 MBytes
9.00-10.00s 6.55 MBytes
41.8 Mbps
68.2 Mbps
66.0 Mbps
81.8 Mbps
39